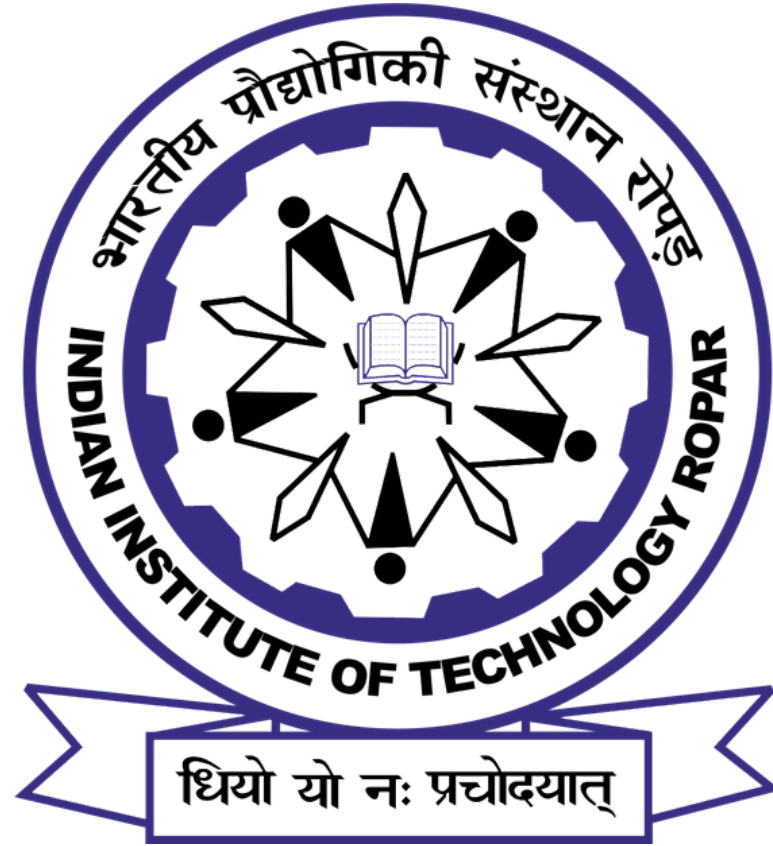




CFD PROJECT

- 1. Lid Driven Cavity Flow**
- 2. Flow Between Parallel Plates**
- 3. Flow Over Flat Plate**

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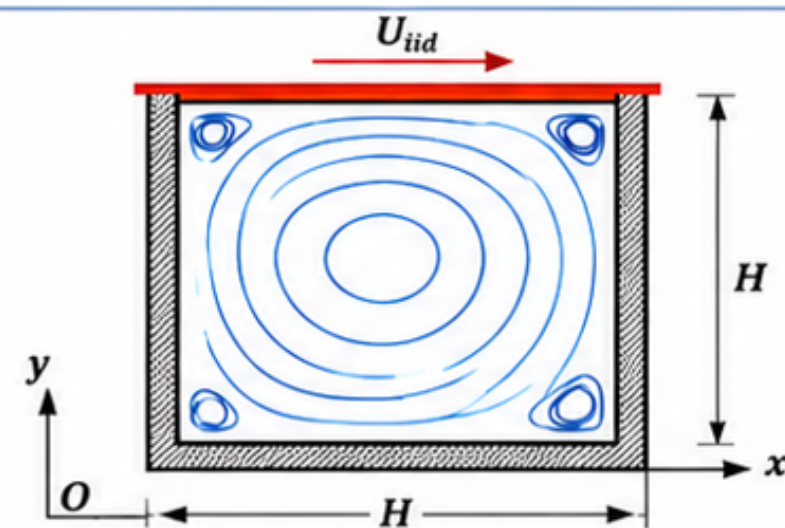
Course Instructor

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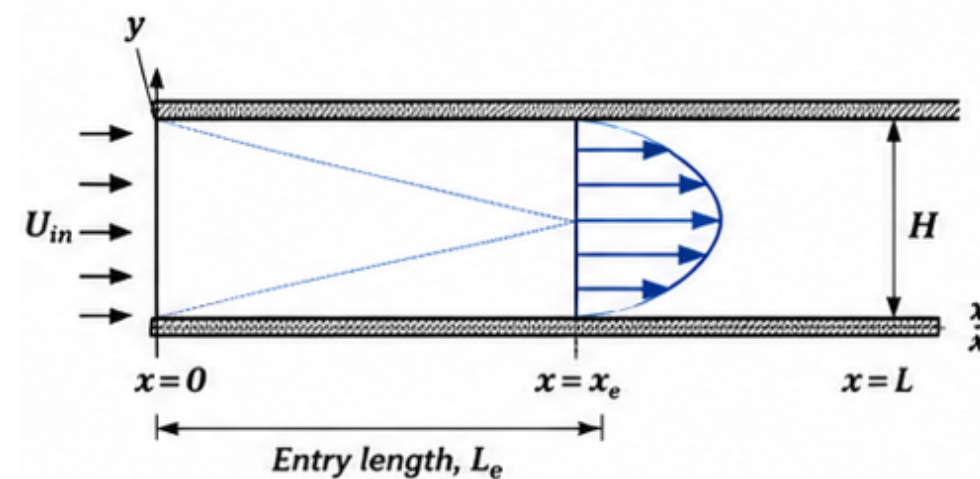
Introduction

- Computational Fluid Dynamics (CFD) is a numerical technique used to analyze and predict fluid flow behavior.
- CFD solves the Navier–Stokes equations using computational methods and numerical discretization techniques.
- In this project, incompressible laminar flow problems are simulated using the Streamfunction–Vorticity formulation and Finite Difference Method.
- The study focuses on internal and external flow behavior, boundary layer development, vortex formation, and Reynolds number effects.

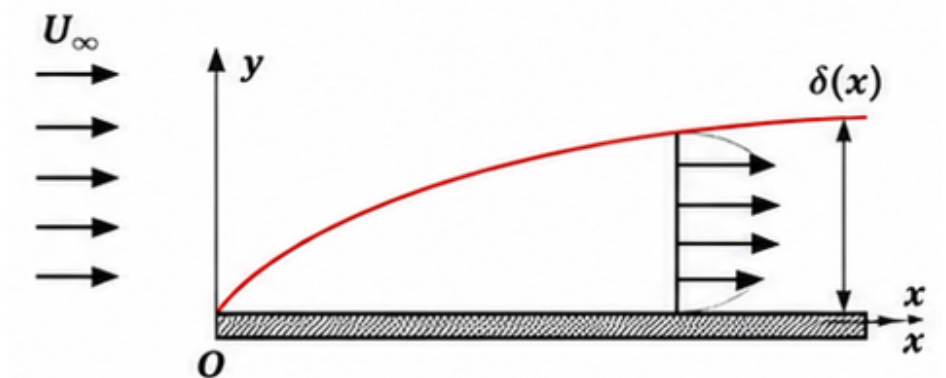
Lid Driven Cavity Flow



Flow Between Parallel Plates



Flow Over Flat Plate



Governing Equations

Governing Equations

Continuity Equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

x-Momentum Equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \nabla^2 u$$

y-Momentum Equation

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \nabla^2 v$$

Assumptions

- Incompressible flow
- Laminar flow
- Newtonian fluid
- Two-dimensional flow

Streamfunction–Vorticity Formulation

Stream Function Relations

$$u = \frac{\partial \psi}{\partial y}$$

$$v = -\frac{\partial \psi}{\partial x}$$

Vorticity Definition

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Poisson Equation

$$\nabla^2 \psi = -\omega$$

Vorticity Transport Equation

$$\frac{\partial \omega}{\partial t} + u \frac{\partial \omega}{\partial x} + v \frac{\partial \omega}{\partial y} = \nu \left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right)$$

Problem 1: Lid-Driven Cavity Flow



DESCRIPTION

- Top wall moves with constant horizontal velocity.
- Remaining walls are stationary.
- Flow develops due to lid motion.
- Recirculation zones and vortices are formed inside the cavity.



BOUNDARY CONDITIONS

Boundary	Velocity Condition	Vorticity Condition
Bottom Wall	$u = 0, v = 0$	$\omega_{bottom} = -2 \frac{\partial \psi}{\partial y}$
Top Lid	$u = 1, v = 0$	$\omega_{top} = -2 \frac{\psi_{n-1}}{dy^2} - \frac{2U}{dy}$
Left Wall	$u = 0, v = 0$	$\omega_{left} = -2 \frac{\partial \psi}{\partial x}$
Right Wall	$u = 0, v = 0$	$\omega_{right} = -2 \frac{\psi_{n-1}}{dx^2}$

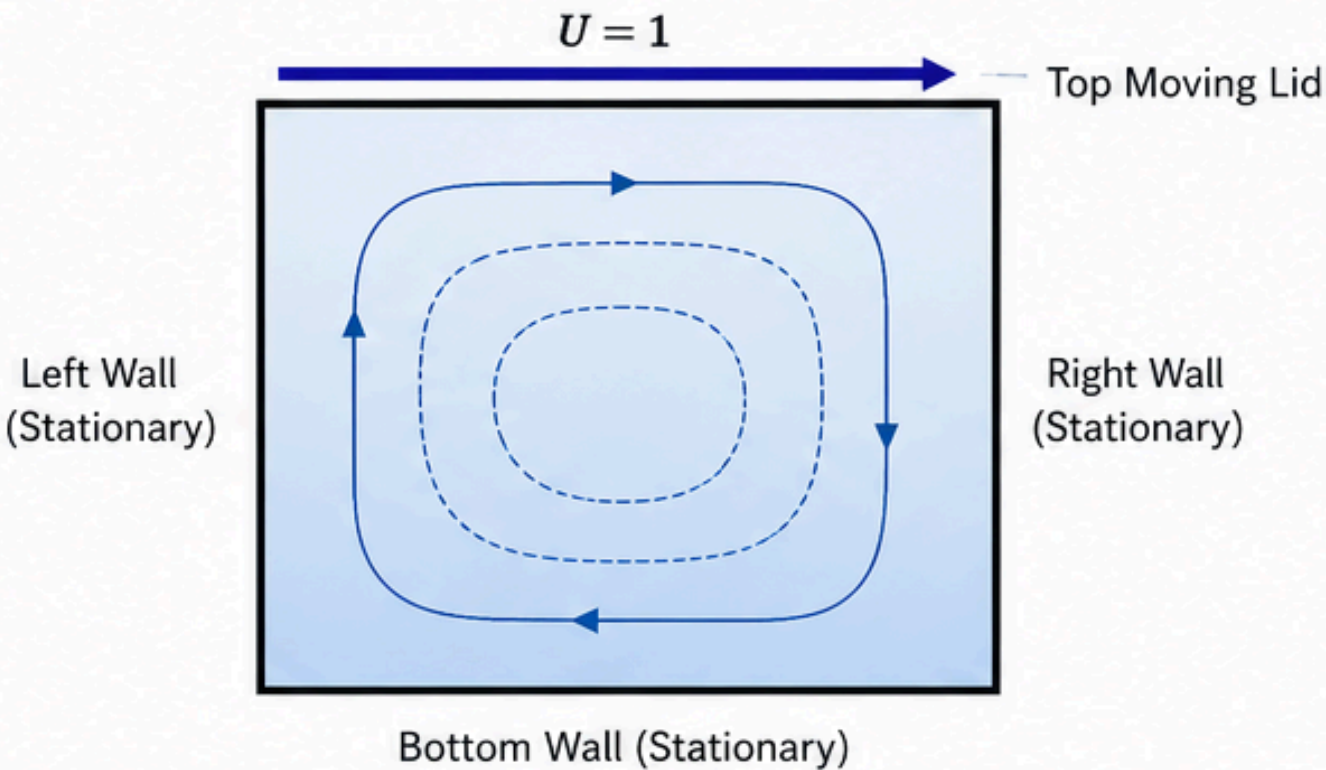


NUMERICAL METHODS AND SCHEMES

Method	Scheme
Spatial Discretization	FDM
First Derivative Terms	Central Difference Scheme
Second Derivative Terms	Central Difference Scheme
Time Marching	Explicit Euler Method
Poisson Solver	Gauss Siedel Method



GEOMETRY (LID-DRIVEN CAVITY)

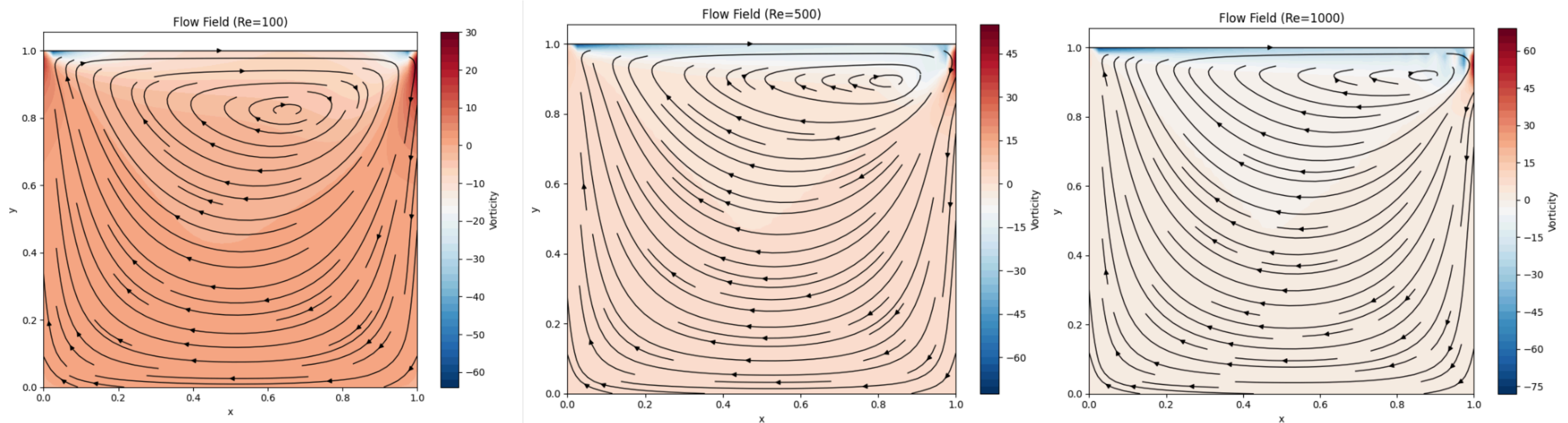


COMPUTATIONAL PARAMETERS

Parameter	Value
Grid Size ($n_x = n_y$)	101 × 101
Domain Length ($L_x = L_y$)	1.0
Grid Spacing (dx)	$\frac{L_x}{n_x - 1}$
Grid Spacing (dy)	$\frac{L_y}{n_y - 1}$
Time Step (dt)	0.0001
Maximum Iterations (nt)	15000
Poisson Iterations (nit)	100
Reynolds Numbers	100, 1000, 500

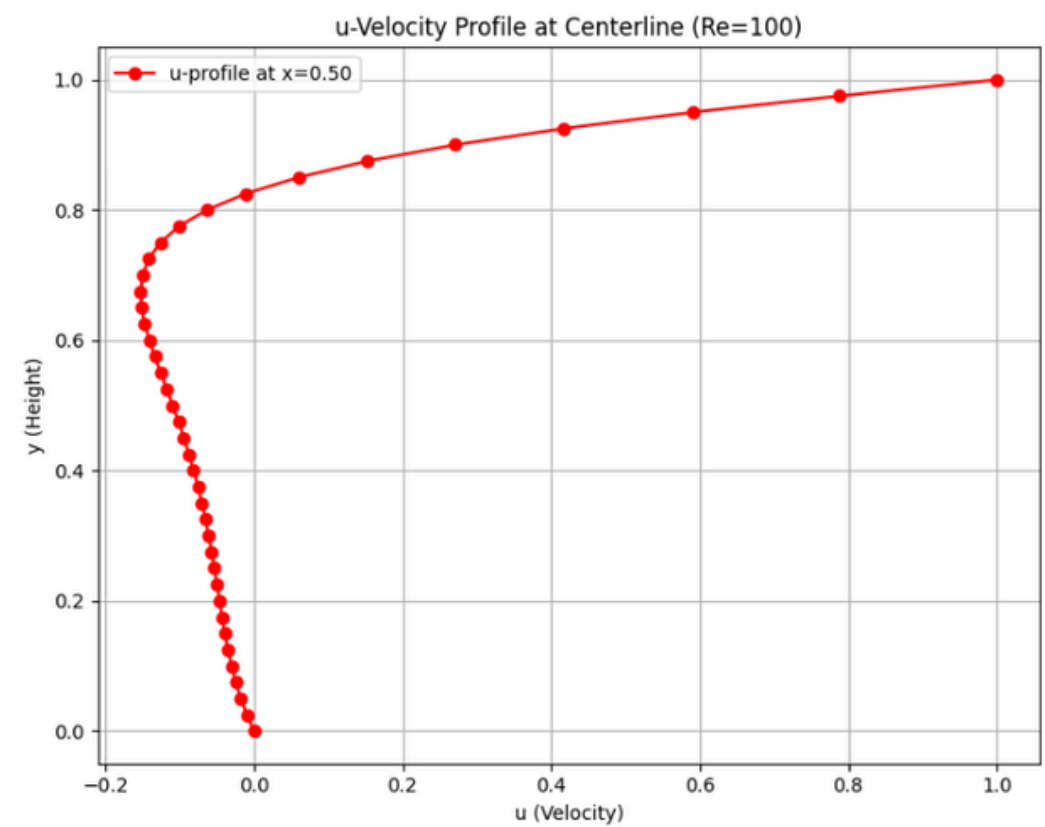
Result

The streamline plots show the flow behavior for different Reynolds numbers ($Re=100, 500, 1000$)
As Reynolds number increases, the vortex strength and circulation inside the cavity increase.
Higher Reynolds numbers produce denser and more curved streamlines near the walls.

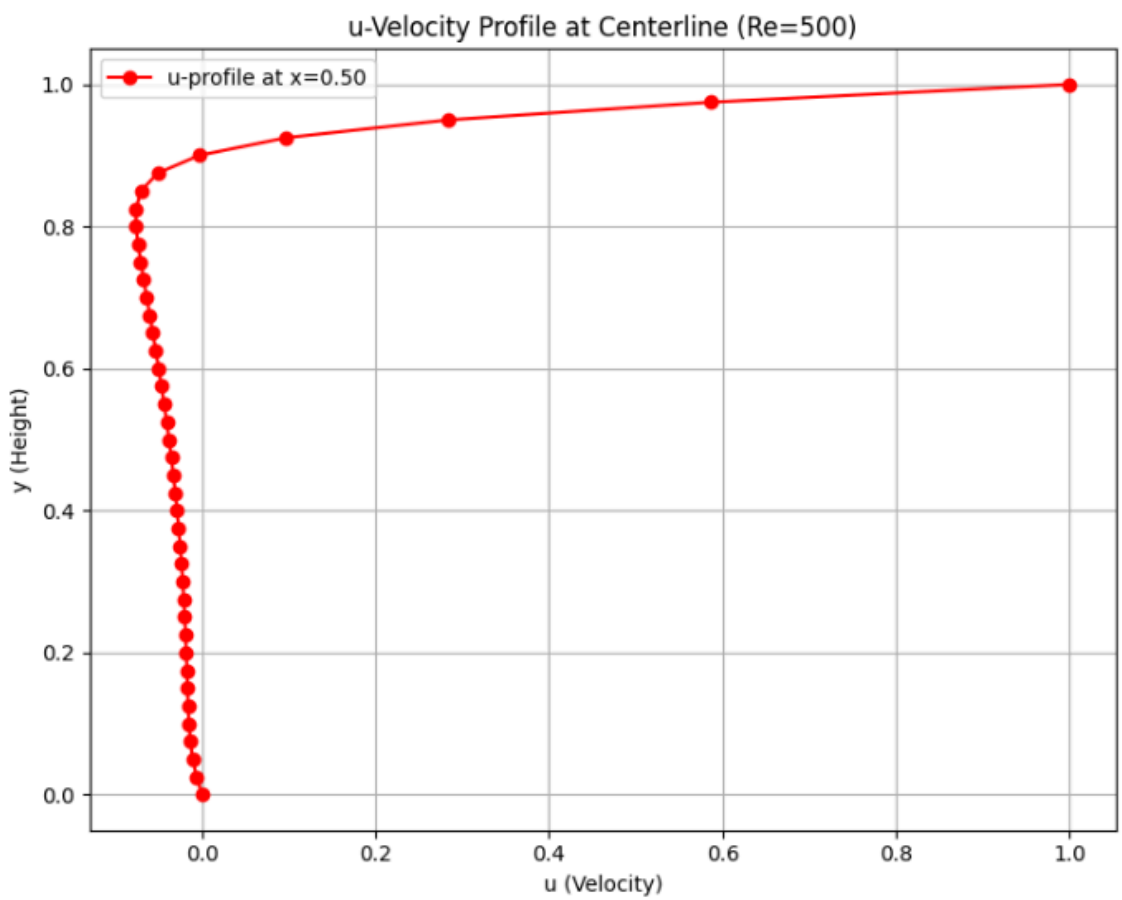


Horizontal Velocity Along Vertical Centreline

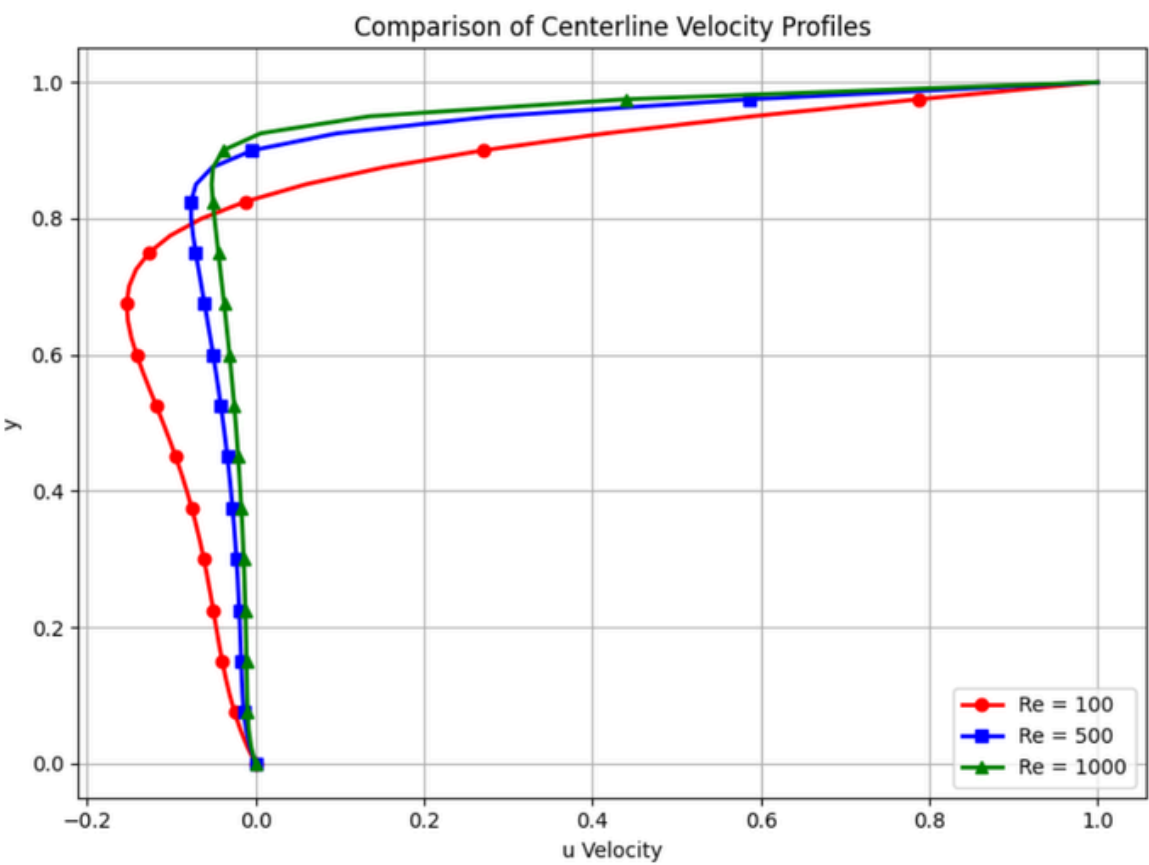
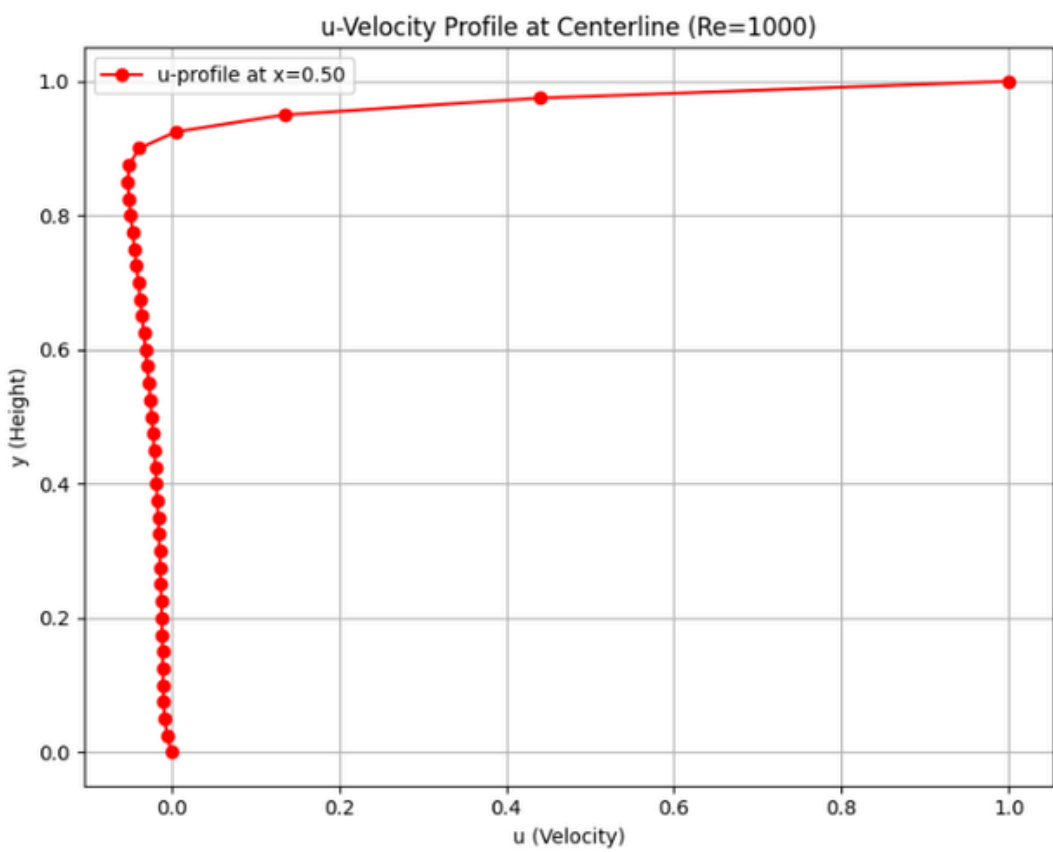
Re 100



Re 500



Re 1000



Poiseuille's flow (or Hagen-Poiseuille flow).

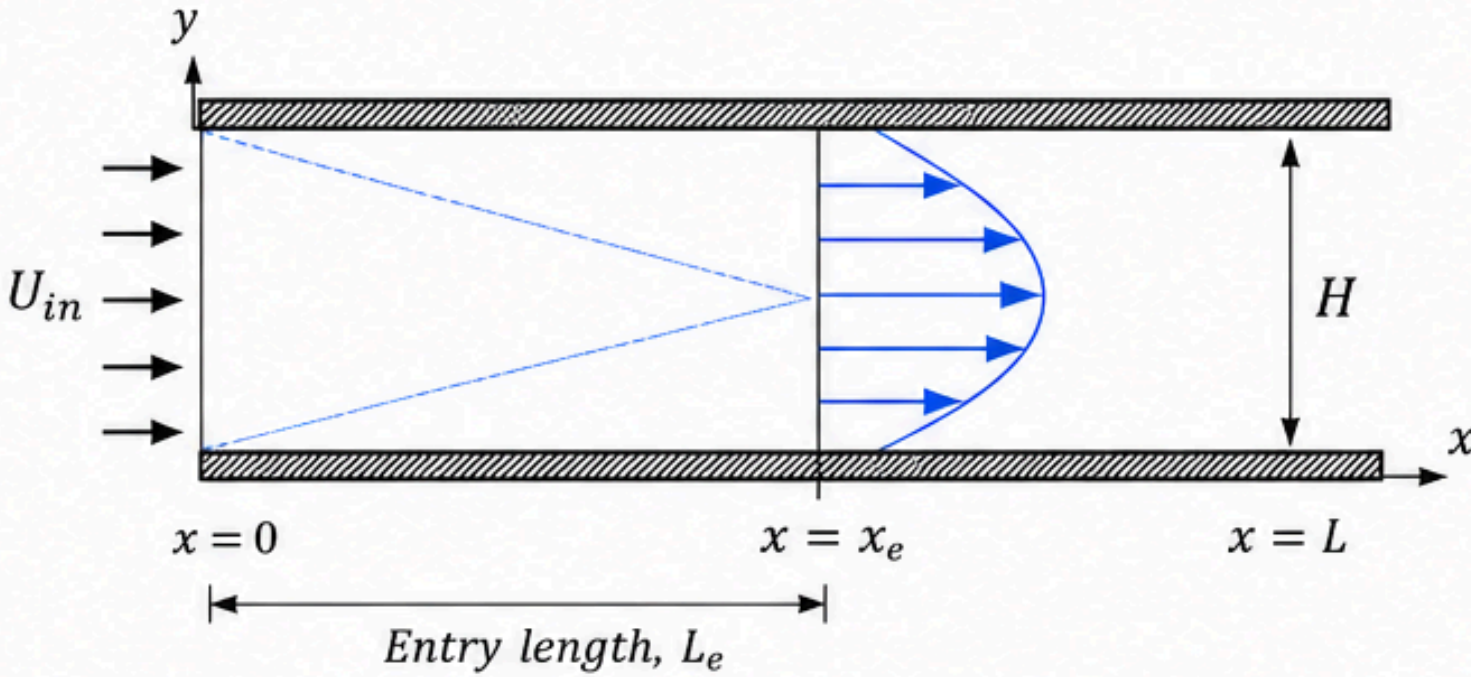


Problem 2: Flow Between Parallel Plates



DESCRIPTION

- Fluid enters a two-dimensional channel formed by two stationary parallel plates with a uniform inlet velocity.
- Due to the no-slip condition at the walls, boundary layers develop from both plates.
- The boundary layers grow downstream and merge at the centerline.
- The velocity profile gradually changes from a uniform profile to a fully developed Poiseuille profile.



BOUNDARY CONDITIONS

Boundary	Velocity Condition	Stream Function Condition	Vorticity Condition
Inlet	$u = U_0, v = 0$	$\psi = U_0 y$	$\omega = 0$
Bottom Plate	$u = 0, v = 0$	$\psi = 0$	$\omega_{bottom} = -2 \frac{\psi_1 - \psi_{wall}}{dy^2}$
Top Plate	$u = 0, v = 0$	$\psi = U_0 H$	$\omega_{top} = -2 \frac{\psi_{N-1} - \psi_{wall}}{dy^2}$
Outlet	$\frac{\partial u}{\partial x} = 0$	$\frac{\partial \psi}{\partial x} = 0$	$\frac{\partial \omega}{\partial x} = 0$



FORMULAE

Reynolds Number

$$Re = \frac{U_0 H}{\nu}$$

Fully Developed Poiseuille Profile

$$u = 6U_0 \left(\frac{y}{H} \right) \left(1 - \frac{y}{H} \right)$$

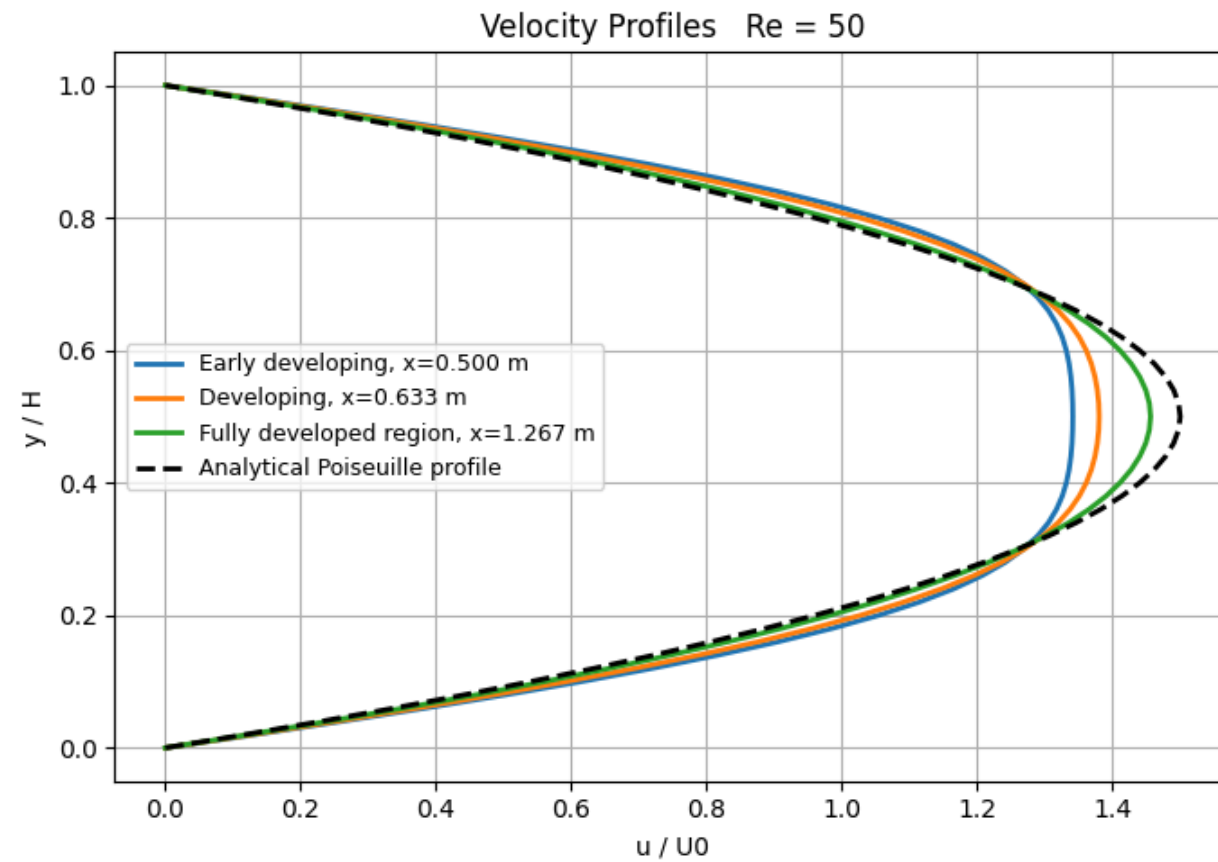
where U_0 = inlet velocity, H = channel height, ν = kinematic viscosity



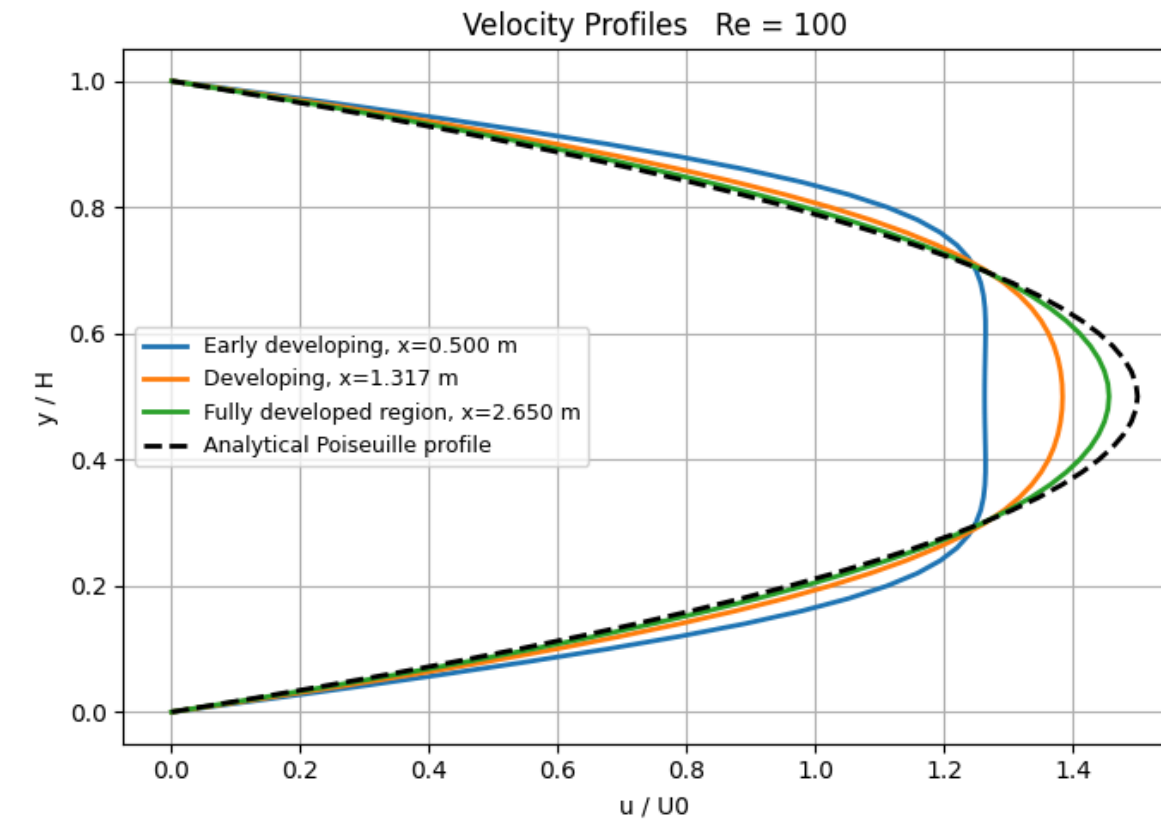
COMPUTATIONAL PARAMETERS

Parameter	Value
Domain Length (L_x)	5.0
Domain Height (L_y)	1.0
Grid Size	301 × 51
Inlet Velocity (U_0)	1.0 m/s
Maximum Iterations	20000
Poisson Iterations	150
Convergence Tolerance	10^{-5}
Reynolds Numbers	50, 100, 200

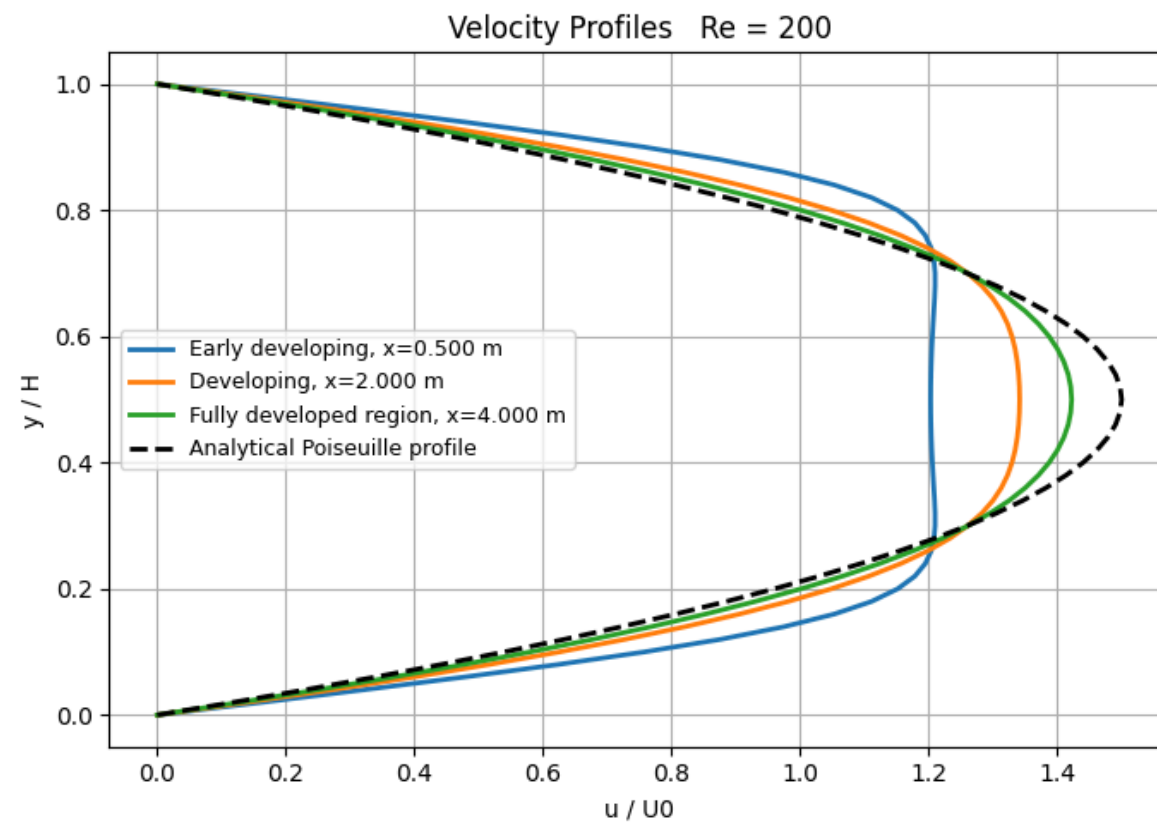
Velocity Profiles at Different Cross-sections



Re 50



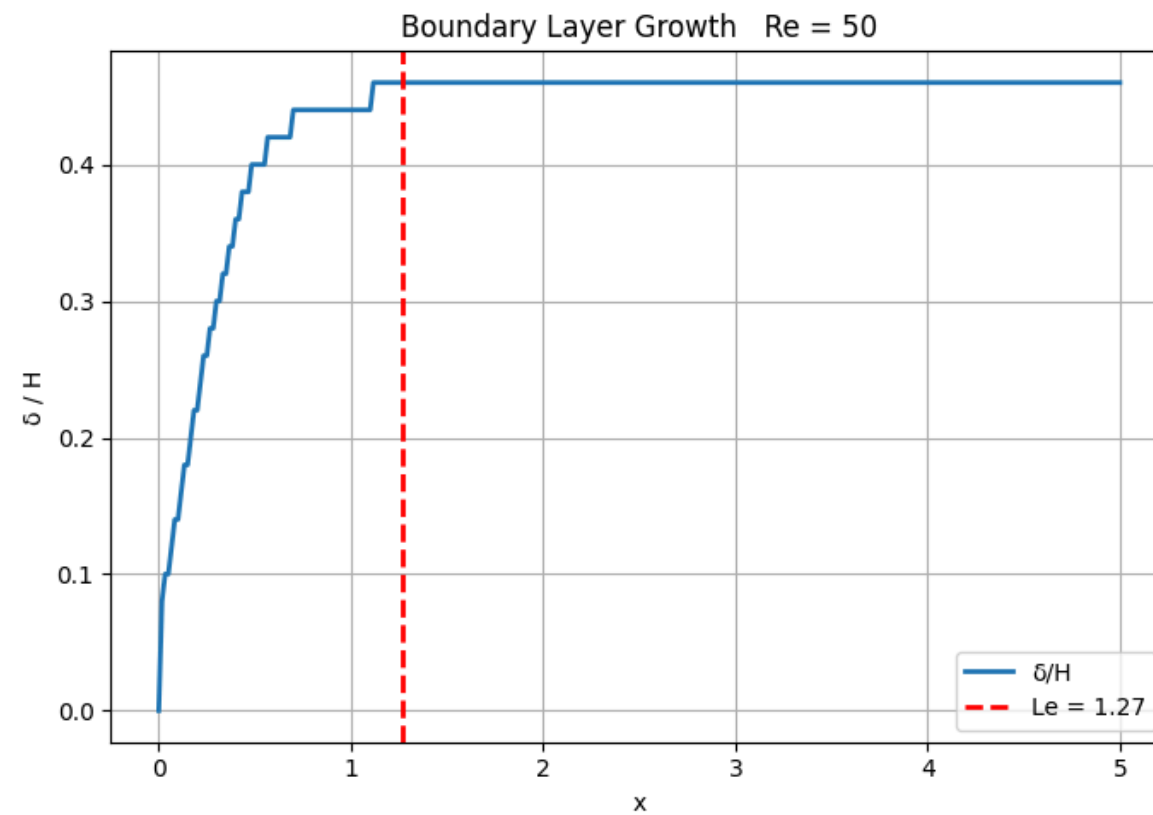
Re 100



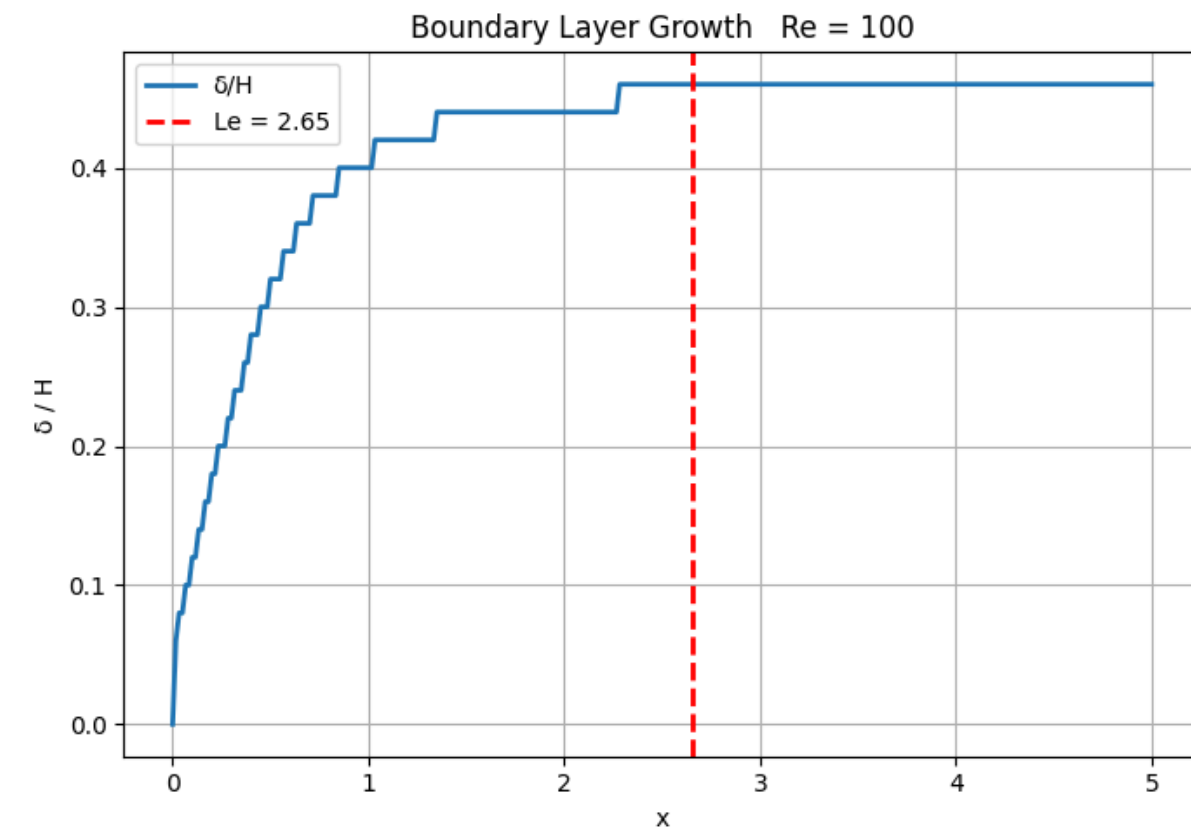
Re 200

1. The inlet velocity profile is initially uniform with constant velocity U_0 .
2. Boundary layers develop near the walls due to the no-slip condition ($u=0$).
3. The velocity profile gradually changes from flat to parabolic downstream.
4. Higher Reynolds numbers delay flow development, increasing the entry length.
5. The numerical profiles converge to the analytical Poiseuille profile, validating the simulation.

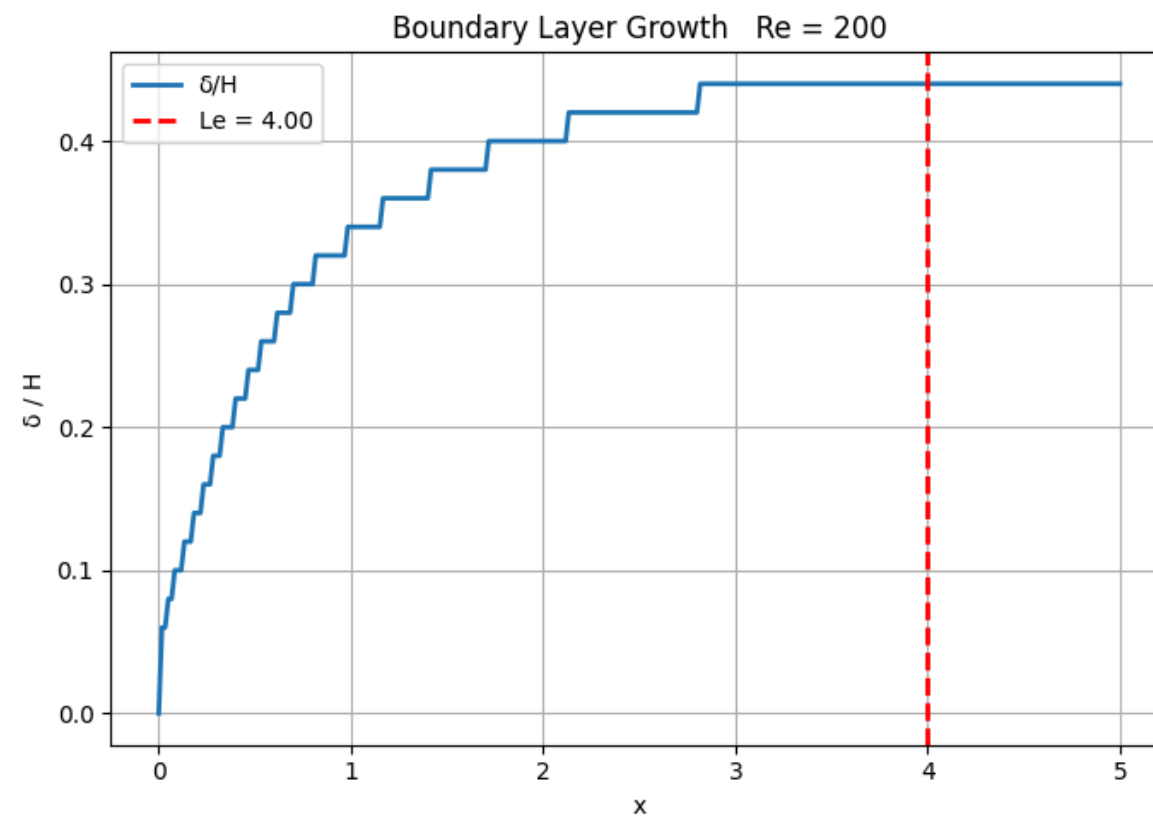
Boundary Layer Thickness V/S Reynold's Number



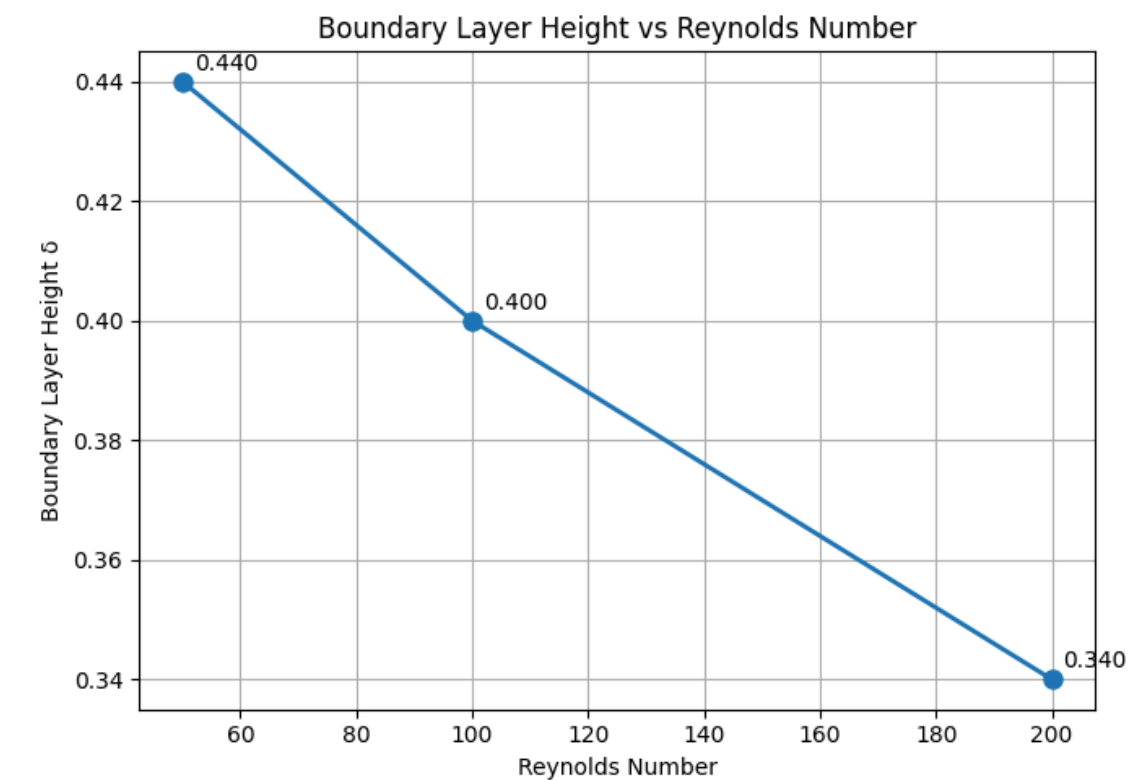
Re 50



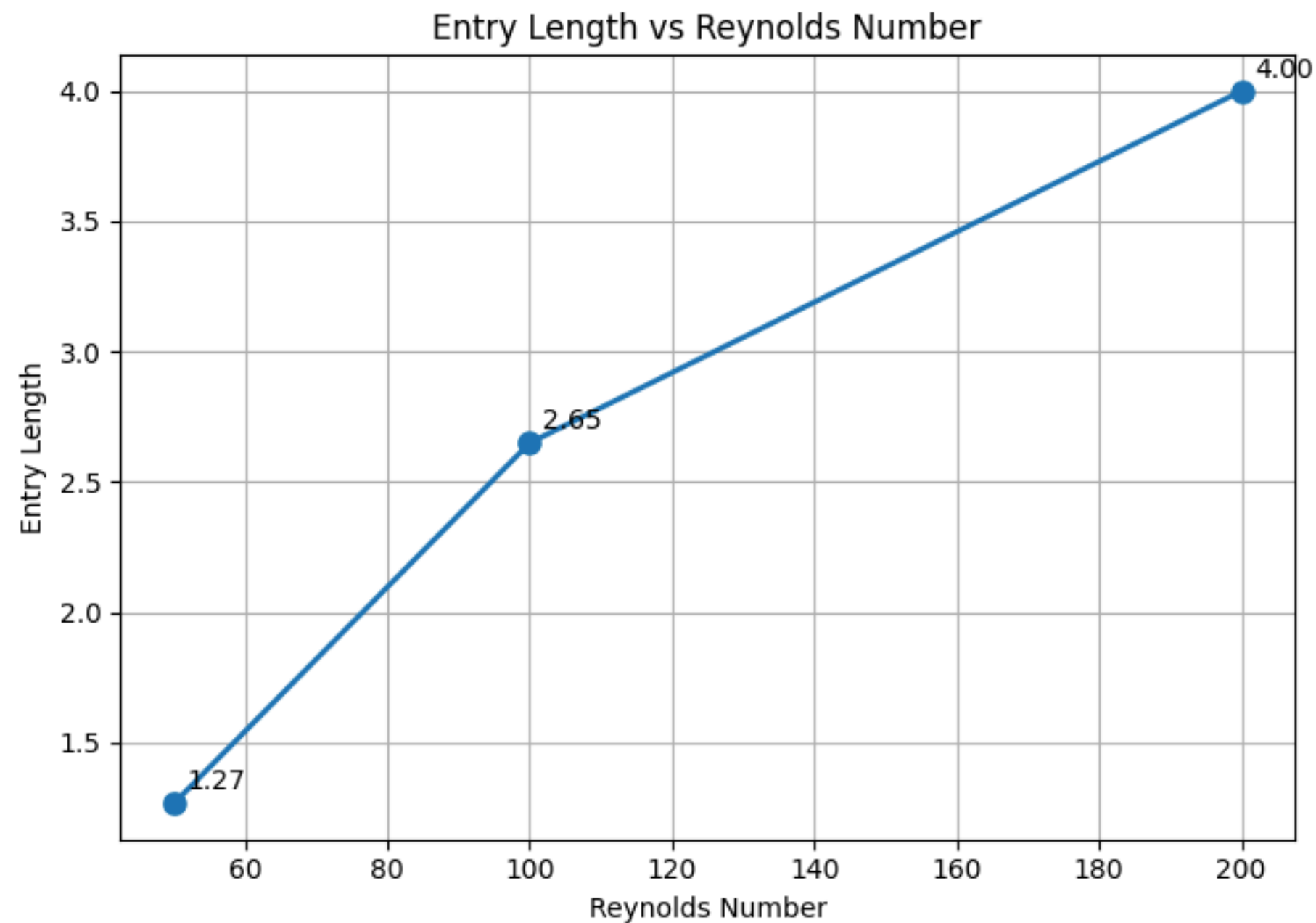
Re 100



Re 200



Entry Length V/S Reynold's Number



Reynolds Number	Entry Length (m)
50	1.27
100	2.6500
200	4.00

- Entry length increases with Reynolds number.
- Low Reynolds number flows develop faster due to stronger viscous effects.
- High Reynolds number flows require longer distances to become fully developed.
- Entry length shows an approximately linear variation with Reynolds number in laminar flow.
- The numerical results match the expected Poiseuille flow behavior, validating the simulation.

Flow over Flat Plate - External Flow

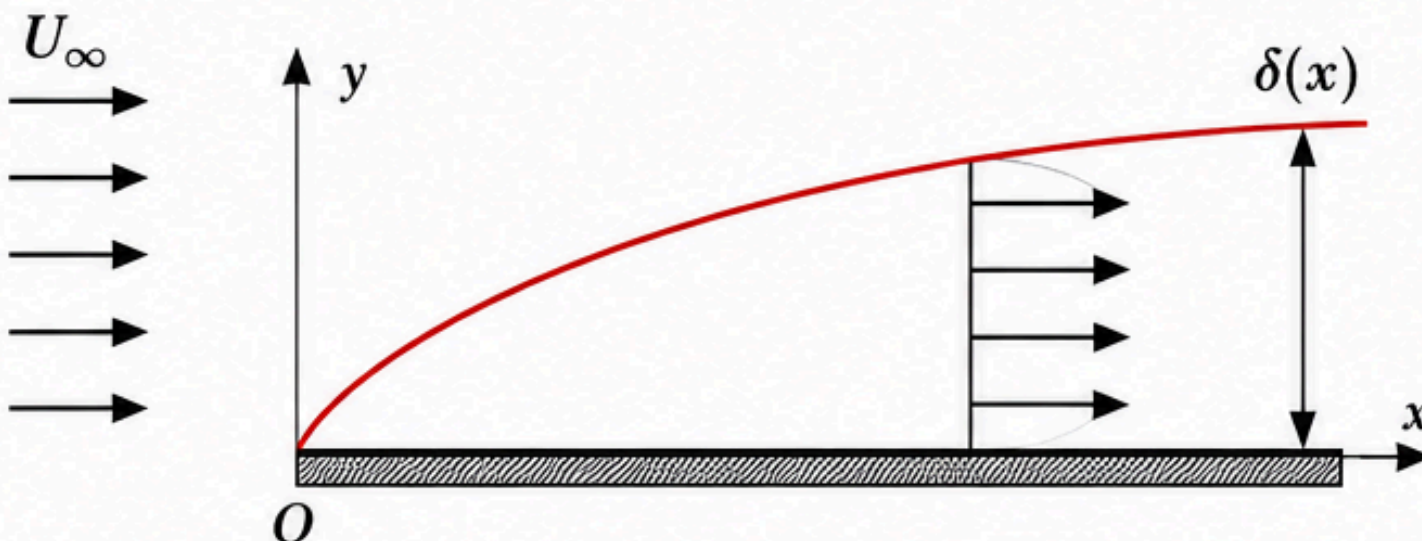


Problem 3: Flow Over Flat Plate



DESCRIPTION

- Uniform freestream flow enters over a stationary flat plate.
- Due to viscosity, a boundary layer develops along the plate.
- Boundary layer thickness increases downstream.



BOUNDARY CONDITIONS

Boundary Section	Velocity Condition	Stream Function Condition	Vorticity Condition
Flat Plate Wall	$u = 0, v = 0$	$\psi = 0$	$\omega_{wall} = -\frac{2\psi_1}{dy^2}$
Inlet Boundary	$u = U_\infty, v = 0$	$\psi = U_\infty y$	$\omega_{inlet} = 0$
Top Boundary	$u = U_\infty, v = 0$	$\psi = U_\infty L_y$	$\omega_{top} = 0$
Outlet Boundary	$\frac{\partial u}{\partial x} = 0$	$\frac{\partial \psi}{\partial x} = 0$	$\frac{\partial \omega}{\partial x} = 0$



FORMULAE

Kinematic Viscosity Relation

$$\nu = \frac{U_\infty L}{Re_L}$$

where
 U_∞ = free stream velocity
 L = characteristic length (plate length)
 Re_L = Reynolds number based on L

Theoretical Boundary Layer Thickness

$$\delta_{99} = \frac{4.91x}{\sqrt{Re_x}}$$

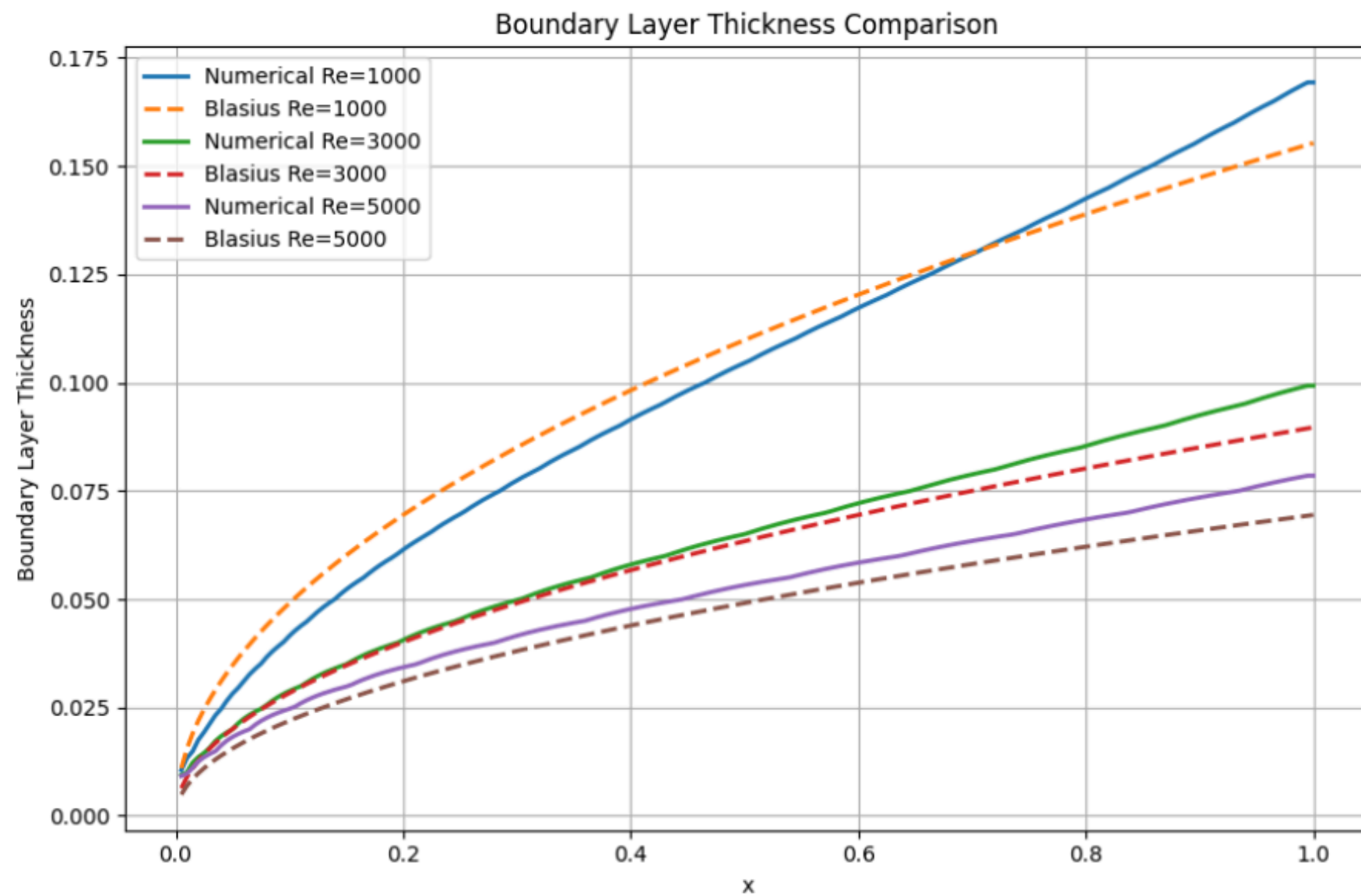
where
 x = distance from leading edge
 $Re_x = \frac{U_\infty x}{\nu}$



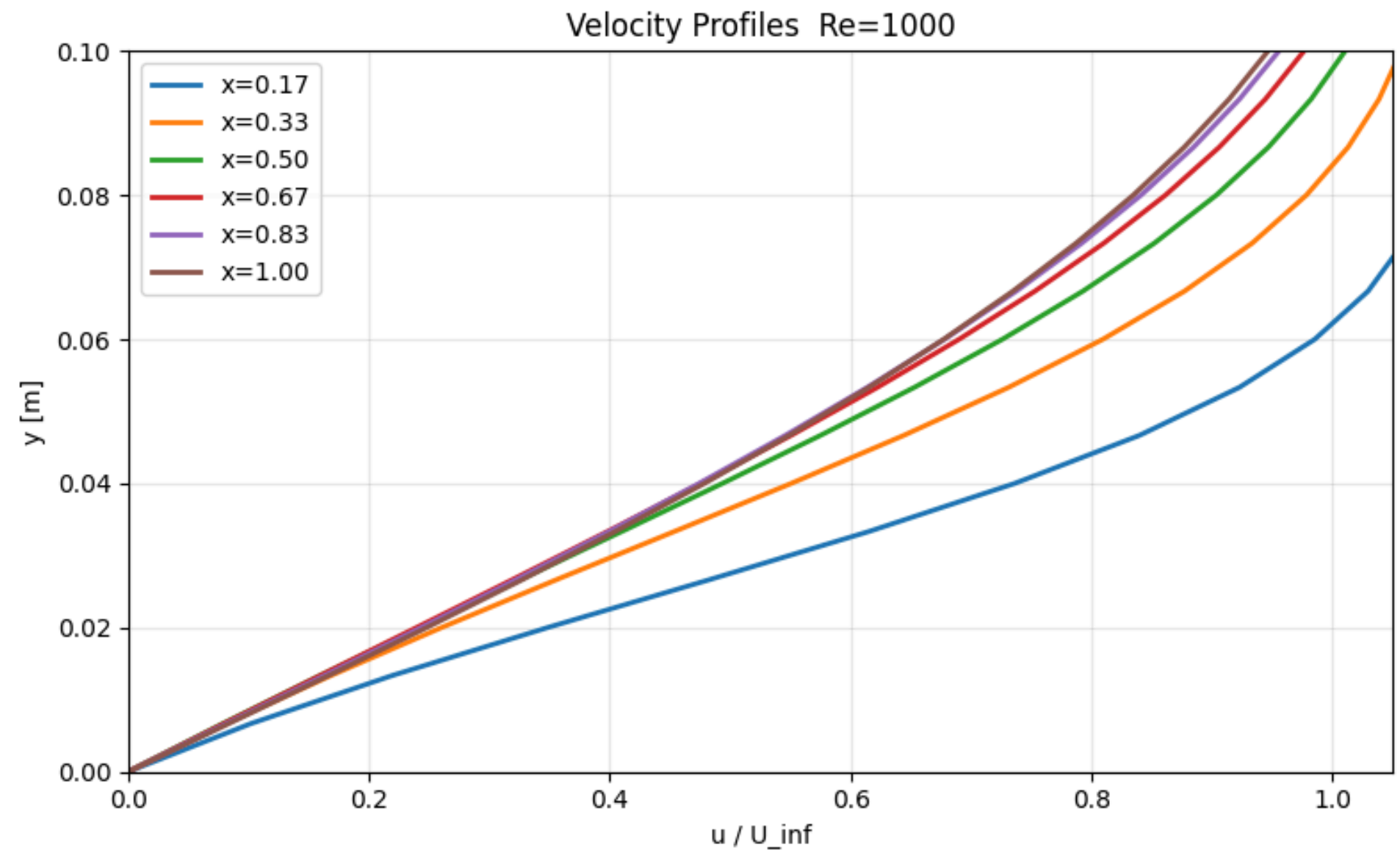
COMPUTATIONAL PARAMETERS

Parameter	Value
Domain Length (L_x)	1.0
Domain Height (L_y)	0.8
Grid Size	121×121
Free Stream Velocity (U_∞)	1.0 m/s
Maximum Iterations	4000
Poisson Iterations	80
Convergence Tolerance	10^{-5}
Reynolds Numbers	1000, 3000, 5000

RESULTS

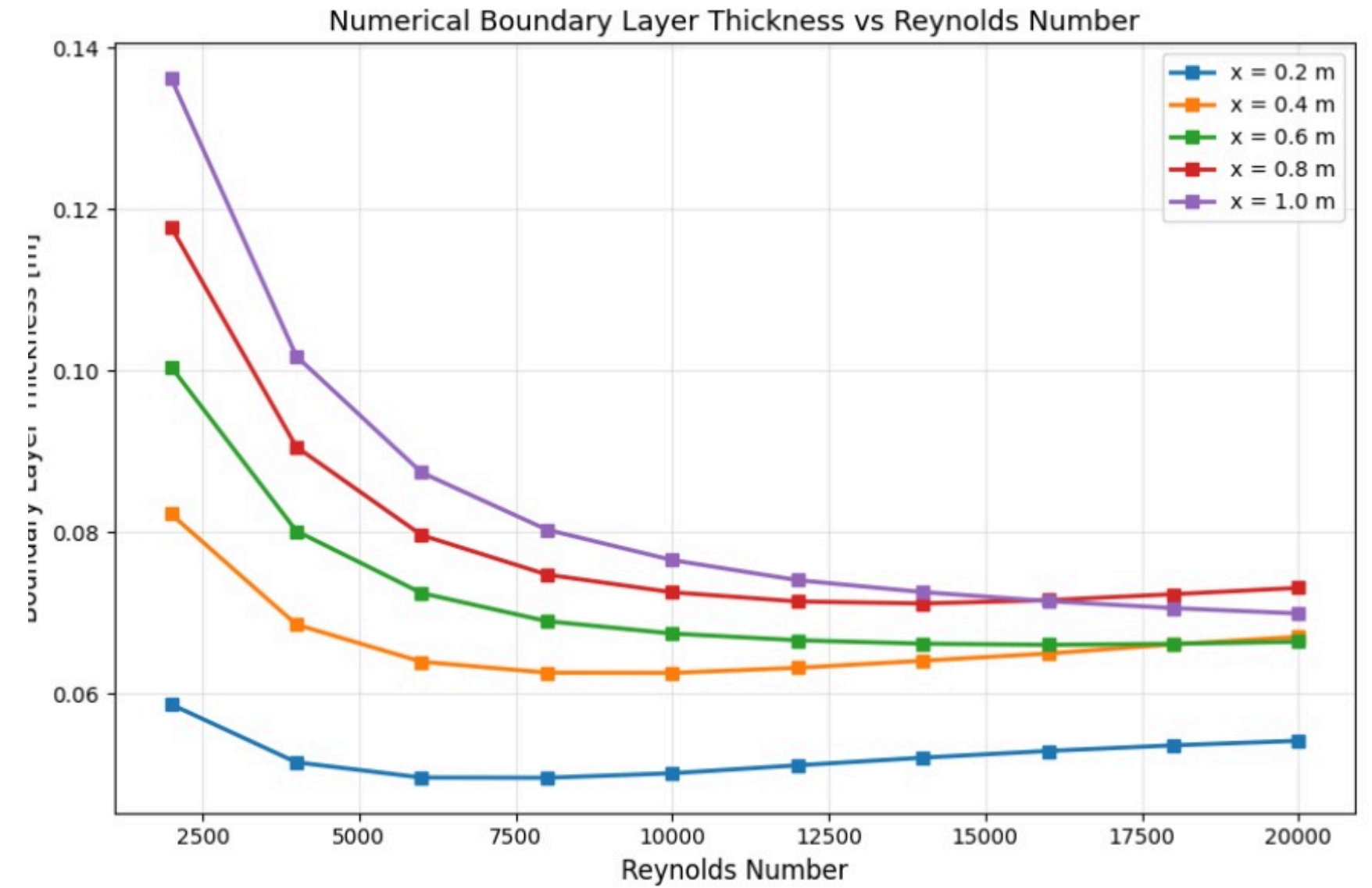
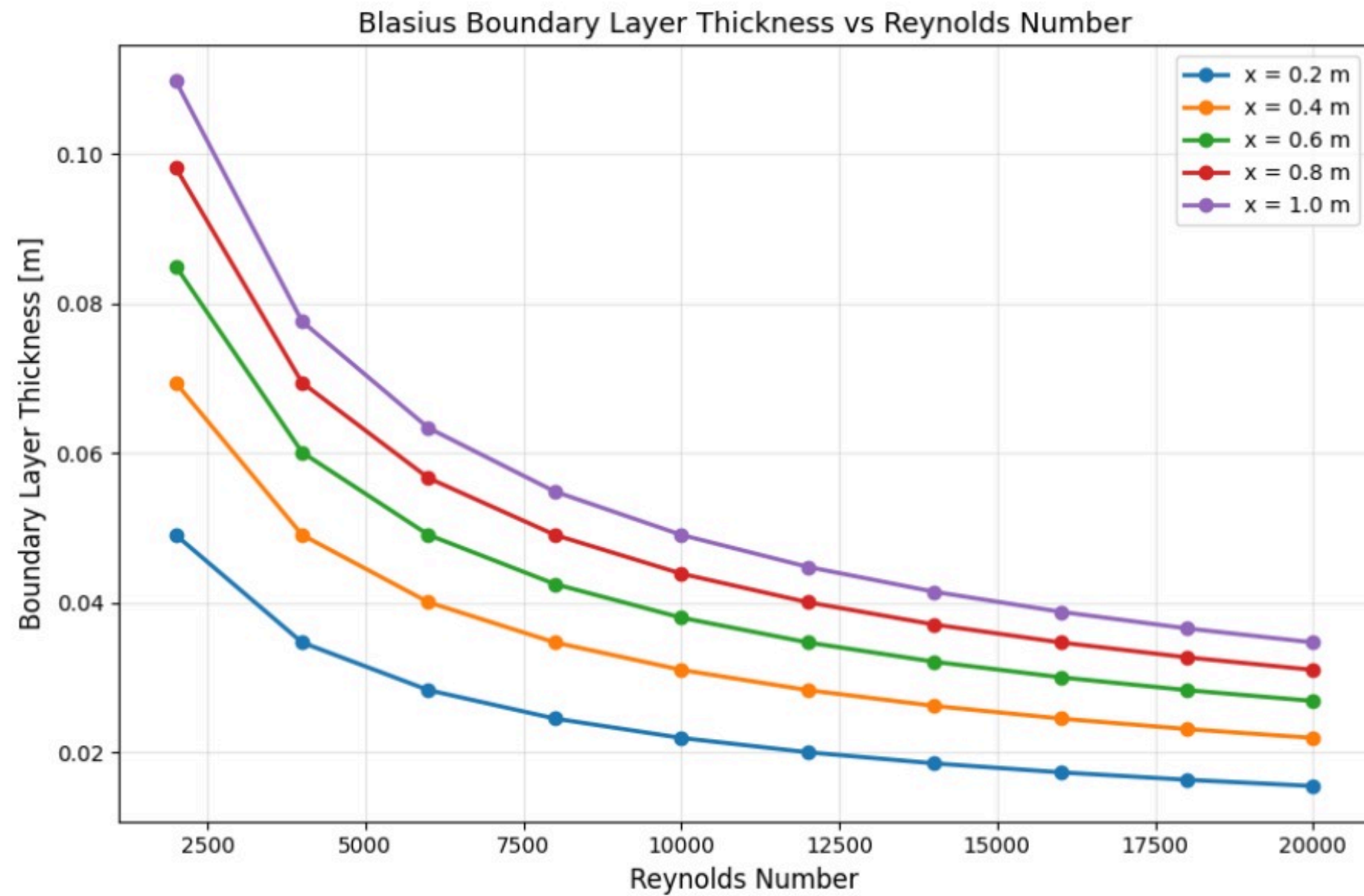


Boundary Layer Thickness vs x



Velocity Profile

Boundary Layer Thickness vs Re (VALIDATION)



Conclusion

- **Lid -Driven Cavity Flow**
- The stream function–vorticity method successfully captured cavity recirculation.
- Increasing Reynolds number strengthened the primary vortex.
- Higher Reynolds numbers produced denser streamlines near the walls.
- Centerline velocity gradients increased with Reynolds number.
- Numerical results matched benchmark cavity flow behavior.

- **Developing Flow Between Parallel Plates**
- Flow developed from a uniform inlet profile to Poiseuille flow.
- Boundary layers grew from both plates and merged downstream.
- Entry length increased with Reynolds number.
- Boundary layer growth followed laminar internal flow theory.
- Numerical results converged to the analytical Poiseuille solution.

- **External Flow Over a Flat Plate**
- Boundary layer thickness increased along the plate length.
- Boundary layer thickness decreased with increasing Reynolds number.
- Velocity profiles gradually approached free-stream velocity away from the wall.
- Numerical results closely matched the Blasius solution.
- FDM successfully captured laminar boundary layer development.